ITDS 124 Mathematics of Computing

Chapter 2. Logic (ตรรกศาสตร์)

2.2 Conditional Statements

(ข้อความที่เป็นเงื่อนไข)

Created by : Rawesak Tanawongsuwan
rawesak.tan@mahidol.ac.th

Conditional statements / Implication (การมีเงื่อนไขทางเดียว)

- If p and q are statement, the compound statement "**if $\boldsymbol{p}$ then $\boldsymbol{q}$**" is denoted by $\boldsymbol{p \rightarrow q}$
  - p is called the **antecedent** or **hypothesis**.
  - q is called the **consequent** or **conclusion**.
- Example :
  - p: I am hungry $\qquad$ q: I will eat

    $p \rightarrow q$: If I am hungry, then I will eat.
  - p: It is snowing $\qquad$ q: $3 + 5 = 8$

    $p \rightarrow q$: If it is snowing, then $3 + 5 = 8$.

Implications in English or language sense

- Example :
  - p: I am hungry. q: I will eat

    $p \rightarrow q$: If I am hungry, then I will eat. ← when we say "if p then q", we assume a cause-and-effect relationship between p and q

  - p: It is snowing. q: $3 + 5 = 8$

    $p \rightarrow q$: If it is snowing, then $3 + 5 = 8$. ← p has no effect on q

Implications in logic

- Implication is used in a much weaker sense. It does not say that p caused q in the usual sense.
- It says $p \rightarrow q$ is true if p is true, then q will also be found to be true.
- Or in other words, $p \rightarrow q$ only says that we will not have p true and q false at the same time.
- The truth table shown here :

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Equivalent expressions of Implication

- Implication $p \rightarrow q$: p implies q

  q if p

  p only if q

  p is a sufficient condition for q

  q is a necessary condition for p

Converse (บทกลับ) and Contrapositive (การแย้งสลับที่)

- The **converse** of $p \to q$ is $q \to p$.
- The **contrapositive** of $p \to q$ is $\sim q \to \sim p$

- Example Let p: It is raining. q: I get wet.
  - Implication $p \to q$: If it is raining, then I get wet.
  - Converse $q \to p$: If I get wet, then it is raining.
  - Contrapositive $\sim q \to \sim p$: If I do not get wet, then it is not raining.

Converse and Contrapositive

p	q	$p \to q$	$q \to p$	$\sim q \to \sim p$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

Equivalence (สมมูล) or Biconditional (เงื่อนไขสองทาง)

- If p and q are statements, the compound statement p if and only if q, denoted by $p \leftrightarrow q$, is called an **equivalence** or **biconditional**.
- $p \leftrightarrow q$ is true only when both p and q are true or when both are false.
- $p \leftrightarrow q$ means p is **a necessary and sufficient condition** for q. (เงื่อนไขที่จำเป็นและเพียงพอ)
- The truth table shown here :

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Proof of the contrapositive

- Compute the truth table of the statement $(p \to q) \leftrightarrow (\sim q \to \sim p)$.

$\mathbf{p}$	q	$p \to q$	$\sim q$	$\sim p$	$\sim q \to \sim p$	$(p \to q) \leftrightarrow (\sim q \to \sim p)$
T	T	T	F	F	T	T
T	F	F	T	F	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	T

Tautology, Contradiction, Contingency

- A statement that is true for all of its propositional variables is called a **tautology** (สัจนิรันดร์).

- A statement that is false for all of its propositional variables is called a **contradiction** (ข้อขัดแย้ง) or an **absurdity** (ความไร้สาระ).

- A statement that can be either true or false depending on truth values of its propositional variables is called a **contingency** (เหตุนิพจน์).

Tautology, Contradiction, Contingency

- Examples
  - $(p \to q) \leftrightarrow (\sim q \to \sim p)$ is a tautology. (true for all cases)
  - $p \wedge \sim p$ is an absurdity. (false for all cases)
  - $(p \to q) \wedge (p \vee q)$ is a contingency.

p	q	$p \to q$	$(p \vee q)$	$(p \to q) \wedge (p \vee q)$
T	T	T	T	T
T	F	F	T	F
F	T	T	T	T
F	F	T	F	F

Logical equivalences (ความสมมูลเชิงตรรกะ)

- Two propositions are logically equivalent or equivalent, if $p \leftrightarrow q$ is a tautology and is denoted by $\boldsymbol{p \equiv q}$.
- Example : The binary operation $\vee$ has the commutative property, $p \vee q \equiv q \vee p$, so $(p \vee q) \leftrightarrow (q \vee p)$ is tautology.

$\boldsymbol{p}$	$\boldsymbol{q}$	$\boldsymbol{p \vee q}$	$\boldsymbol{q \vee p}$	$\boldsymbol{(p \vee q) \leftrightarrow (q \vee p)}$
T	T	T	T	T
T	F	T	T	T
F	T	T	T	T
F	F	F	F	T

Logically equivalent

- Example. We can look for columns in the truth table for each statement and compare them to see if they are identical. $p \to q$ and $(\sim p) \vee q$ are equivalent.

p	q	$p \to q$	$\sim p$	$(\sim p) \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Theorem 1 Properties of logical operations

- Commutative Properties
  - $p \vee q \equiv q \vee p$
  - $p \wedge q \equiv q \wedge p$
- Associative Properties
  - $p \vee (q \vee r) \equiv (p \vee q) \vee r$
  - $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$
- Distributive Properties
  - $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
  - $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

Theorem 1 Properties of logical operations

- Idempotent Properties
  - $p \vee p \equiv p$
  - $p \wedge q \equiv p$
- Properties of Negation
  - $\sim(\sim p) \equiv p$
  - $\sim(p \vee p) \equiv (\sim p) \wedge (\sim q)$ De Morgan's law
  - $\sim(p \wedge p) \equiv (\sim p) \vee (\sim q)$ De Morgan's law

Theorem 2 Properties of implication operation

- Properties of implication operation
  - $(p \to q) \equiv \big((\sim p) \vee q\big)$
  - $(p \to q) \equiv (\sim q \to \sim p)$
  - $(p \leftrightarrow q) \equiv ((p \to q) \wedge (q \to p))$
  - $\sim(p \to q) \equiv (p \wedge \sim q)$
  - $\sim(p \leftrightarrow q) \equiv ((p \wedge \sim q) \vee (q \wedge \sim p))$

Theorem 3 : Properties for the ∀ and ∃ quantifiers

- Properties for the ∀ and ∃ quantifiers
  - $\sim\big(\forall x\ \mathrm{P}(x)\big) \equiv \exists x \sim\mathrm{P}(x)$
  - $\sim\big(\exists x\ \mathrm{P}(x)\big) \equiv \forall x \sim\mathrm{P}(x)$
  - $\exists x\ \big(\mathrm{P}(x) \to \mathrm{Q}(x)\big) \equiv \forall x\ \mathrm{P}(x) \to \exists x\ \mathrm{Q}(x)$
  - $\exists x\ \big(\mathrm{P}(x) \vee \mathrm{Q}(x)\big) \equiv \exists x\ \mathrm{P}(x) \vee \exists x\ \mathrm{Q}(x)$
  - $\forall x\ \big(\mathrm{P}(x) \wedge \mathrm{Q}(x)\big) \equiv \forall x\ \mathrm{P}(x) \wedge \forall x\ \mathrm{Q}(x)$
  - $\big(\big(\forall x\ \mathrm{P}(x)\big) \vee \big(\forall x\ \mathrm{Q}(x)\big) \to \forall x(\mathrm{P}(x) \vee \mathrm{Q}(x))$ is a tautology.
  - $\exists x\big(\mathrm{P}(x) \wedge \mathrm{Q}(x)\big) \to \exists x\ \mathrm{P}(x) \wedge \exists x\ \mathrm{Q}(x)$ is a tautology.

Theorem 4 : tautologies that are implications

- Tautologies that are implications
  - $(p \wedge q) \rightarrow p$
  - $p \rightarrow (p \vee q)$
  - $\sim p \rightarrow (p \rightarrow q)$
  - $\big(p \wedge (p \rightarrow q)\big) \rightarrow q$
  - $\big(\sim q \wedge (p \rightarrow q)\big) \rightarrow \sim p$
  - $(p \wedge q) \rightarrow q$
  - $q \rightarrow (p \vee q)$
  - $\sim(p \rightarrow q) \rightarrow p$
  - $\big(\sim p \wedge (p \vee q)\big) \rightarrow q$
  - $\big((p \rightarrow q) \wedge (q \rightarrow r)\big) \rightarrow (p \rightarrow r)$